

README - HW6 - Image Processing

Introduction

In this assignment, we are making an image-processing application that has many image-processing effects. This assignment will be written in java using some designs such as MVC, function object design, and command pattern.

Parts Have Been Completed

Everything in the hw5 still works perfectly fine. We thought we had a bug when running the script file in the jar; however, it is now working since I removed the old jar file and built the new one. Other than that, we have done everything for this assignment, which is creating a GUI for user interaction. Users now could load the image that they like in their folder. (When loading the bpm file, it would have a delay, and the users could click any effect it will show up the image). Users now could do all the functionalities in the previous assignments through the view. They also can do the command lines and run in the JAR file and also see the histogram that represents the distribution of color or intensity.

Changes We Made

We added a new class for the View (ImageGuiView), implementing the interface IView, a new controller to work with the GUI, a MockGuiView to test the GuiController, and a Histogram class. We also change the old controller to become an abstract class to remove the code duplication between the text-UI controller and the GUI controller. All other aspects of the project remain unchanged.

Purpose and Design

For the design in this assignment, we are still using the MVC model. Basically, we still keep the same purpose and design from the last assignment but this time we add the GUI. Since the purpose and design are the same in README homework 5, we only add the changes for this section

Main Class

- We run the program through this main class, and with the command lines, the users would have the option to run a script file, type the script, and open the GUI
- Changes: added functionality to the command line so that it can be access through the terminal and calling the text-UI or Gui or just run a script file.

Controller Package

- Changes: This package now has an ImageGUIController, which will help the user interact and do all the functionalities that the users do while using the app. This class also interacts with the view and the model, making sure that the view and model will not interact with each other, which is the MVC design we follow. And to remove code duplication, we changed the base of the old controller to become an abstract controller

and have the controller extends it. This might help if any futures controller demands any changes.

- This class implements the IController, and the IViewListener to do the viewEventPerfome, which is a function doing all the effects when they receive the String of command.

Functions Package and Model Package

- We do not make any changes to this.

Folder res

- This folder will have an image.ppm, image.png, image.jpg, and image.bmp which serve the small test in the controller test.
- It has the jar file which can be accessed to run the program from the terminal.
- It has the script.txt file which contains the valid command to execute in our program.
- Script2.txt contains the command for sepia and grayscale which are failing right now. But the function works when used in the main, and controller, etc.
- It has the Diagram.png which is the diagram of the current project.
- It also has the example images required by the assignment:
 - + Citation: These are the pictures of Lam's dog, which are taken by our phones, so there is no problem using the pictures! The submitted images are compressed by compress2go.com.
 - + Dog1.jpg is the example of the jpg file
 - + Dog.png is the example of the png file
 - + Dog2.bmp is the example of the bmp file
 - + Image2.ppm is an example of the ppm file. We tried to convert our picture to ppm, but it is too large, so we decided to include a 5x5 pixel picture for the example of ppm.
 - + Other examples demonstrate the effect reflected by its name.
 - + We included the screenshot of our program with a image loaded named GUIHW6.png

View Package

- In this package, we will have a histogram class. This class shows the histogram of the app, which is a way to visualize the distribution of color or intensity. We also have an ImageGuiView class, implementing the ActionListener and IView interface.
- The ImageGuiView class will handle nearly everything related to UI stuff, which is showing the interactive view for the users. Users could now do every functionality that the ImageProcessing app has with this view.
- This class will have the actionPerformed function that will take action events and do things accordingly.